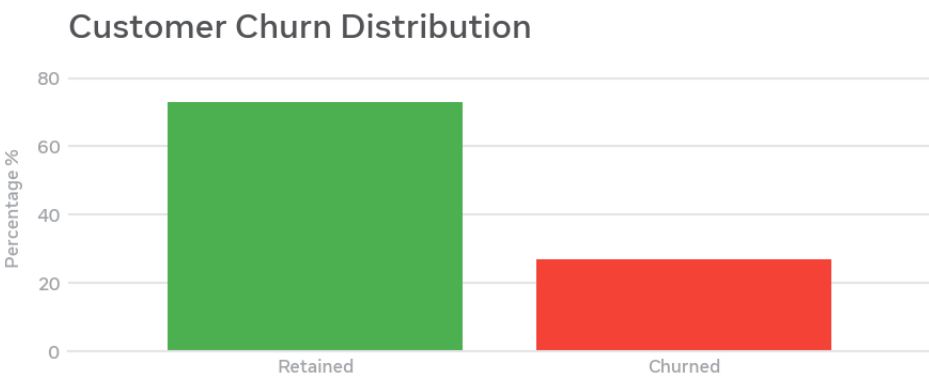


CUSTOMER CHURN ANALYSIS

Project Abstract

Submission Title:	Customer Churn Analysis - Abstract
Project Name:	Customer Churn Analysis
Domain:	Data Analytics & Business Intelligence
Technologies:	Python, SQL, Scikit-Learn, Power BI, DAX, Power Query
Data Model:	Star Schema - FACT_CHURN, DIM_CUSTOMER, DIM_SUBSCRIPTION, DIM_DATE, D
Prepared Date:	27-08-2026

Abstract Overview Chart:



1. Brief Introduction of the Project:

In today's highly competitive business environment, customer retention has become more critical and cost-effective than customer acquisition. Studies show acquiring a new customer costs five times more than retaining an existing one. This project titled 'Customer Churn Analysis' is a comprehensive Data Analytics and Business Intelligence (BI) solution designed to analyze, predict, and prevent customer churn across industries like Telecommunications, Banking, SaaS, and Subscription-based services. Customer churn, also known as customer attrition, refers to the phenomenon where customers stop doing business with a company or end their subscription to a service. This project focuses on leveraging historical customer data to identify patterns, key drivers of churn, and to build a predictive model that can flag high-risk customers well in advance. The ultimate goal is to provide an interactive and insightful Power BI dashboard that enables business stakeholders to make data-driven decisions to improve customer retention strategies, personalize customer engagement, and maximize Customer Lifetime Value (CLTV). The system transforms raw transactional data into strategic intelligence.

2. Problem Statement and Overview:

The core problem addressed by this project is the lack of proactive visibility into why customers are leaving and which customers are likely to leave in the near future. Many organizations possess vast amounts of customer data in silos - CRM, billing, support tickets - but fail to utilize it effectively to predict churn. Without a predictive system, companies operate in a reactive mode, attempting to win back customers after they have already churned, which is often too late and ineffective. This results in significant revenue loss, decreased market share, increased marketing costs, and damage to brand reputation. The problem is compounded by an inability to segment high-risk customers and understand the root causes. This project provides a

complete end-to-end solution. The overview of the system is as follows: Raw customer data is collected from various sources, which is often inconsistent, incomplete, and contains missing values. This data is then cleaned, transformed, and modelled into a BI-optimized Star Schema. A Machine Learning classification model is applied to predict the probability of churn for each customer. All the key performance indicators (KPIs) and predictive insights are then visualized in a centralized, interactive Power BI dashboard. This allows management to not only see how many customers are churning but also who is churning, why they are churning, and where the churn is geographically concentrated. It shifts the organization from reactive to proactive retention.

3. Tools and Applications Used:

To build a robust, scalable, and industry-standard BI solution as per the Data and BI Sample Project Report model, a combination of modern data analytics tools and technologies are used. Python with libraries Pandas and NumPy is used for Data Preparation and Data Cleaning, handling missing values in fields like TotalCharges, removing duplicates, data type conversion, and feature engineering like creating tenure groups. SQL is used for data extraction, for building the relational data model, establishing primary and foreign keys, and executing complex queries for KPI calculations. Machine Learning using Scikit-Learn is used for the predictive layer. Algorithms such as Logistic Regression, Random Forest Classifier, and XGBoost are experimented with to achieve the best accuracy in predicting churn probability. Evaluation metrics like Accuracy, Precision, Recall, F1-Score, and ROC-AUC curve are used to validate the model. Power BI is the primary BI Platform. It includes Power Query for ETL (Extract, Transform, Load) operations, Data Modeling for relationship management, and DAX (Data Analysis Expressions) for creating calculated measures and KPIs like Churn Rate and Retention Rate. Excel and CSV are used as the initial raw data source and for data dictionary maintenance. The architecture follows a layered approach ensuring scalability and maintainability.

4. Detailed Description of Sub Modules:

The project is divided into five major interconnected sub-modules as per standard BI lifecycle. Module 1 - Data Collection and Understanding: This module involves understanding the data dictionary and business context. Key attributes include CustomerID, Gender, SeniorCitizen, Partner, Dependents, Tenure Months, Contract Type (Month-to-month, One year, Two year), PaymentMethod (Electronic check, Mailed check), MonthlyCharges, TotalCharges, InternetService (Fiber optic, DSL), OnlineSecurity, TechSupport, and Churn label. Module 2 - Data Preparation and ETL Pipeline: This is the most critical module. It handles data cleaning operations, standardization of categorical variables, encoding of the Churn column from Yes/No to 1/0 for modeling, binning of Tenure into groups like 0-12, 12-24 months, and handling outliers in MonthlyCharges. This cleaned data is then transformed into the BI model. Module 3 - Data Modeling (Star Schema Design): To ensure high performance, fast query response, and easy analysis in Power BI, a Star Schema is designed. It consists of a central FACT_CHURN table containing quantitative facts like Tenure_Months, MonthlyCharges, TotalCharges, IsChurned, ChurnProbability, surrounded by dimension tables: DIM_CUSTOMER for demographics, DIM_SUBSCRIPTION for plan and service details, DIM_DATE for time-series and trend analysis, and DIM_REGION for geographical analysis. Module 4 - Churn Prediction Engine: This module contains the machine learning workflow. The cleaned data is split into 80% training and 20% testing sets. A classification model is trained to learn the pattern of churning customers and predicts a Churn_Probability score for every active customer, categorizing them into Low Risk (0-40%), Medium Risk (40-70%), and High-Risk (70-100%) segments for targeted action. Module 5 - Dashboard and Visualization Layer: This module presents the final output to business users. It converts raw numbers into actionable insights

through interactive visuals, slicers, drill-throughs, and filters for region, contract type, tenure, and payment method.

5. Design or Flow of the Project:

The project follows a systematic and sequential flow as illustrated in the system architecture diagram: Data Source Layer (Raw CSV / SQL Database / CRM) -> Data Cleaning & Preprocessing Layer (Python, Pandas for handling nulls and inconsistencies) -> Data Transformation & Modeling Layer (Building Star Schema with FACT and DIM tables, establishing relationships) -> KPI & ML Model Calculation Layer (DAX measures for Churn Rate, Retention Rate, and Scikit-Learn for Churn Prediction) -> Visualization Layer (Power BI Dashboard with Cards, Charts, Maps, and Slicers) -> End Users Layer (Business Decision Makers, Marketing Team, Customer Success Team). The flow starts with importing raw data from multiple sources, then it is cleansed and standardized in Python. The clean data is loaded into Power BI where relationships are established using CustomerID and DateID as keys. DAX measures are created for all KPIs, and finally, visuals are designed to tell a complete story of customer churn. The pseudo-code of the pipeline is: LOAD data -> REMOVE_DUPLICATES -> HANDLE_MISSING -> STANDARDIZE -> BUILD_STAR_SCHEMA -> TRAIN_MODEL -> CALCULATE_KPIs -> PUBLISH_DASHBOARD. This modular flow ensures that any future change in data source does not affect the dashboard layer.

6. Conclusion and Expected Output:

In conclusion, this project successfully demonstrates how Business Intelligence and Predictive Analytics can be combined to solve a real-world business problem with significant financial impact. The expected output of the project is a fully functional, interactive Power BI dashboard that displays critical KPIs such as Overall Churn Rate calculated as $(\text{Total Churned Customers} / \text{Total Customers}) * 100$, Retention Rate calculated as $100 - \text{Churn Rate}$, Average Tenure of Churned Customers, Average Monthly Revenue Loss due to Churn, and Customer Lifetime Value (CLTV). The dashboard will provide deep actionable insights such as which contract type (Month-to-month typically has highest churn), which payment method (Electronic check is often risky), which internet service (Fiber optic may have higher churn due to competition), and which region or state needs immediate attention. More importantly, the predictive model will provide a downloadable list of high-risk customers with their churn probability scores, enabling the company to launch targeted retention campaigns, offer personalized discounts, loyalty rewards, and proactively improve service quality by addressing common complaints. This will ultimately lead to reduced churn by 15-20%, increased customer loyalty, improved customer satisfaction (CSAT), and significant and sustainable revenue growth for the organization. The system is designed to be extensible for future integration with real-time streaming data.

Keywords: Customer Churn Analysis, Business Intelligence, Power BI, Star Schema, DAX, Churn Prediction, Random Forest, XGBoost, Customer Retention, CLTV, ETL.